



Modeling of a DC-DC bidirectional converter used in mild hybrid electric vehicles from measurements

Gabriel Rojas-Dueñas, Jordi Roger Riba^{*}, Manuel Moreno-Eguilaz

Universitat Politècnica de Catalunya, Electrical Engineering Department, Rambla Sant Nebridi 22, 08222 Terrassa, Catalunya, Spain

ARTICLE INFO

Keywords:

DC-DC bidirectional converter
Mild hybrid electric vehicle
Deep learning
Modeling
Neural network

ABSTRACT

This paper presents a non-intrusive approach for modeling a bidirectional DC-DC converter used in mild hybrid electric vehicles. A black-box identification methodology is proposed to find a model based on the data acquired from the input/output terminals. Measured data include the steady state and transient response, and different operating conditions of the DC-DC converter, including the buck and boost modes. A deep learning architecture based on a long-short-term memory neural network (LSTM-NN) is applied. The trained network is tested under a set of operating points different from those used during the training stage. The proposed method is compared with three black-box modeling techniques commonly used in power converters, proving its superior performance. Results presented in this paper indicate that the proposed model is able to replicate the behavior of the bidirectional converter without a priori knowledge of the converter circuitry. This approach can also be applied to other power devices.

1. Introduction

Over the last years, the automotive industry has been moving towards more efficient vehicles in order to achieve stringent CO₂ emission targets [1]. To achieve this goal automobiles must be electrified, and in particular the powertrains [2,3]. Although full electric vehicles are possibly the best solution because they allow the highest reduction in emission of pollutants [2], a wide expansion is not yet possible since they are expensive compared to conventional ones and the lack of required infrastructures. Therefore, hybrid electric vehicles arise as a solution for the transition from the fuel based technology to the full electric vehicles.

Hybrid electric vehicles (HEVs) consist of two or more power sources, in which one of them is characterized by being electrical and reversible [3]. This paper analyzes a mild hybrid electric vehicle (MHEV), this technology generating around 15 % less emissions than conventional vehicles [4]. MHEVs reduce the fuel consumption by using regenerative braking and by moving the operating points of the thermal engine to more efficient areas [5] with the help of the electrical motor. The deceleration energy is stored in an electrical storage system [6], which uses a 48 V –to- 12 V dual-voltage architecture consisting of two batteries and a bidirectional DC-DC power converter [7]. In most cases,

automobile manufacturers acquire these electrical components from external suppliers. Manufacturers of electrical components often do not provide enough information on the dynamics, operating points or topology of their products [8]. This presents a challenge for MHEV engineers because they require accurate models of the electrical equipment for designing, optimization, control, fault diagnosis [9] and useful life estimation purposes. Thus, it is important to model and identify these elements to fulfill the aforementioned requirements. Black-box models can be useful in addressing this problem, as they do not require prior knowledge of the topology of the device to be identified or modeled. Different authors have proposed modeling approaches of the 48 V and 12 V batteries used in HEVs [6,10]. However, just some few studies have analyzed the identification of bidirectional DC-DC converters. Therefore, this paper attempts to fill this gap, as it seeks to obtain an accurate black-box model of the 48 V –to- 12 V bidirectional power converter used in MHEVs based on an offline black-box identification methodology that uses a deep learning neural network architecture.

Depending on the information available, different approaches can be applied to model power converters. A white-box parameter identification approach is used when the topology of the DC-DC converter is known, although the parameter values are unknown. These problems are usually solved by solving the analytical equations describing the

^{*} Corresponding author at: Universitat Politècnica de Catalunya, Electrical Engineering Department, Rambla Sant Nebridi 22, 08222 Terrassa, Catalunya, Spain.
E-mail addresses: gabriel.esteban.rojas@upc.edu (G. Rojas-Dueñas), jordi.riba-ruiz@upc.edu (J. Roger Riba), manuel.moreno.eguilaz@upc.edu (M. Moreno-Eguilaz).

dynamics of the circuit and using optimization methods [11]. Alternatively, black-box modeling methods, as in this paper, have a higher complexity due to the lack of information about the physical characteristics of the power converter. In this case, the only available information are the currents and voltages at the input and output terminals [12]. These modeling approaches prevent manufacturers of disclosing the information of their products [13].

In [14] different black-box modeling techniques of power converters are reviewed, including the Hammerstein and Steiglitz IIR models, which show a good performance. However, these are online estimation approaches, requiring an external change of the duty cycle, leading to a higher complexity and limited robustness because these methods require a specific model for each operating point [15]. In [16] the frequency response of the power converter is obtained, which is used to generate a transfer function able to replicate its behavior. Despite of allowing to replicate the dynamics of the converter, this method might be inaccurate when the converter works in the discontinuous conduction mode. In [12] a polytopic model is proposed, which obtains the G-parameters of the two-port network (converter) and calculates the linear response at a certain operating point. This method presents accurate results, but it is limited to the analyzed conduction mode of the DC-DC converter, thus failing to estimate changes between the continuous and discontinuous conduction modes. In [17] a black-box model of a buck converter is obtained by using nonlinear autoregressive exogenous neural networks (NARX-NNs). An offline identification is performed and the results show a good resemblance between the estimation and the measured output of the converter. Nevertheless, the computational burden is high and the inputs of the black-box are not the same as the ones of the converter. In [18] an identification method based on recursive neural networks is presented, however, it fails to replicate with high-fidelity the behavior of the converter and it just uses data gathered from simulation. The authors of [19] propose a dynamic system identification method of a DC-DC bidirectional converter used in automobiles. It is based on three different techniques, each one being applied in a certain condition of the converter. It uses the state space averaging method, which is also applied in [20] to obtain the model of a buck converter. Although the error obtained is very low, it was just tested under buck mode, and the operating point of the converter must be known in order to apply one of the three techniques. In [9] the authors use neural networks for predictive control and fault diagnosis of power converters, respectively. However, a model that replicates the behavior of the converter is not obtained.

The methodology proposed in this study is based on offline and non-intrusive measurements of the DC-DC bidirectional converter. Some of them are used to train a long short-term memory neural network (LSTM-NN), while the others test the accuracy of it. An experimental setup capable of obtaining the needed datasets is implemented. It automatically acquires different conditions of the power converter in order to provide enough information to the neural network to generate an adequate model. One of the main advantages of using this deep learning approach is that the neural network is able to learn the long term dependencies of the converter behavior as well as to predict the future outputs based on the past and actual states of the LSTM-NN. Another advantage is the capacity of replicating the steady-state and transient response characteristics of the converter using only one model, and not several, as in other works. Thus, different operating points of the converter can be replicated by means of a single trained LSTM-NN. It is worth noting that the only needed measurements are non-intrusive and they are made at the input and output terminals of the converter, which prevents an alteration of the inner elements of the converter. Also, unlike other studies [18], the experimental setup allows a robust modeling of the bidirectional DC-DC converter. The entire MHEV electrical system can be simulated if an existing battery model [21,22] is added to the bidirectional converter.

Section 2 details the methodology proposed in this paper as well as the electrical system of MHEVs. Section 3 details the LSTM-NN dealt

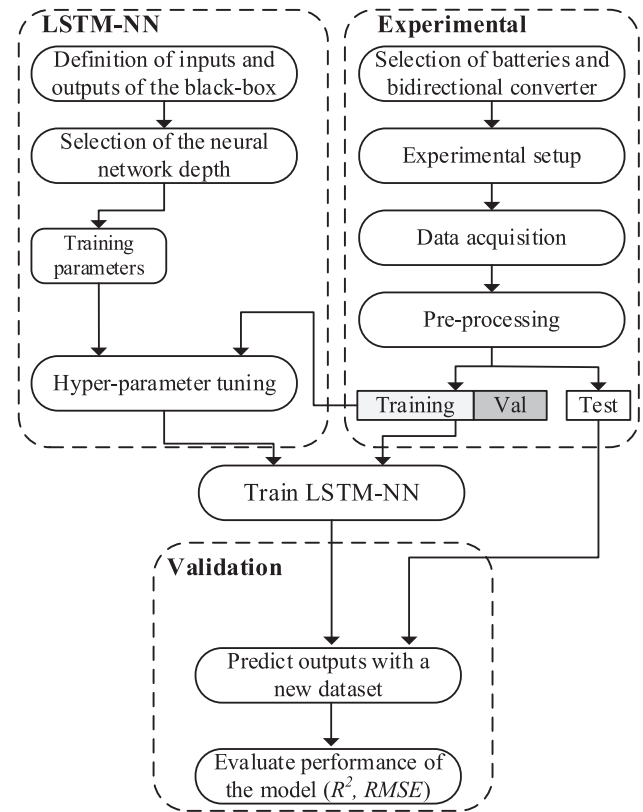


Fig. 1. Proposed measurement-based methodology.

with in this work, including the suggested design and the tuning of the hyper-parameters of the neural network. In Section 4, the experimental setup and measurement strategy are presented and in Section 5 the attained results are shown and analyzed. Finally, Section 6 concludes the paper.

2. Methodology

This paper uses a deep learning approach to model a 48 V –to– 12 V DC-DC bidirectional converter used in mild hybrid electric vehicles. The proposed methodology considers three stages, which comprise the acquisition of the experimental data from the converter, the design and implementation of the LSTM-NN and the validation of the model.

Fig. 1 shows the general procedure to acquire the experimental data, train and validate the model, where the three stages are clearly defined.

The first stage of the proposed methodology comprises the data acquisition of the bidirectional DC-DC converter. To this end, it is necessary to define the equipment to be used and the data to be measured in the laboratory. An automatic data acquisition system was programmed for this purpose, which is able to store a predefined number of experiments. The gathered data is resized and filtered to increase the model accuracy and decrease the training time. Finally, the resulting dataset is divided in the training, validation and test sets. A more detailed explanation of the experimental stage is given in Section 5.

The LSTM-NN stage shown in Fig. 1 defines the neural network and selects the training parameters. First, the inputs and outputs of the NN are specified, in this case corresponding to the measurements at the input and output terminals (voltages and currents) of the bidirectional converter. Next, the number of hidden layers, the fixed parameters and the hyper-parameters to be tuned are defined. In the last step of this stage, the hyper-parameters are tuned by means of an optimization algorithm, which is further explained in Subsection 3.2. Next, the LSTM-NN is trained using the optimized hyper-parameters and the training and validation datasets. To determine the accuracy of the model, the

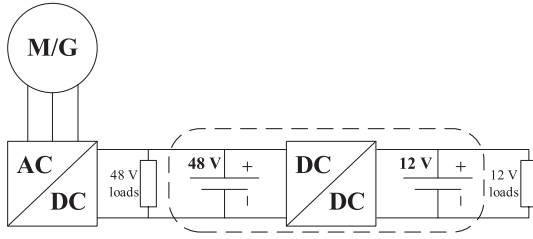


Fig. 2. MHEV electrical system.

neural network is tested under new conditions, and the predicted outputs are compared to the actual measured outputs of the bidirectional converter. Finally, the root mean squared error (RMSE) and the coefficient of determination (R^2) for the two predicted signals are calculated.

2.1. Mild hybrid electrical vehicle (MHEV)

The system to be evaluated in this paper is a MHEV, this topology being used by major manufacturers such as Audi, Toyota, Nissan or Honda, among others [23]. This type of vehicle improves fuel efficiency by supplying power to the electric motor in certain situations or by applying regenerative braking. MHEVs require a higher voltage (48 V) to feed the electrical systems of the vehicle in order to reduce conduction losses [5,24]. The MHEV is a transition technology that is placed in the middle point between the conventional 12 V supply system and the 48 V system proposed by some manufacturers of premier cars [7].

This paper focuses on the dual-voltage architecture that considers two batteries and a DC-DC bidirectional converter, the last one being the device to be modeled. Understanding the working principle and the topology of the electrical system of the MHEV is fundamental in order to obtain an accurate model of the power converter.

Fig. 2 shows the dual architecture of this electrical system and its main components. It can be seen that at the 48 V side there is a battery, a starter generator, a drive (DC-AC or AC-DC depending whether the machine works as a motor or as a generator, respectively) and loads, whereas at the 12 V side there is a battery and a set of low voltage loads. Fig. 2 does not specify the topology of the DC-DC converter since it may change depending on the manufacturer of the automobile and its requirements [7]. However, for the purpose of this paper this is indifferent, since the proposed black-box approach is able to model the bidirectional converter despite of its topology.

Each battery is in charge of stabilizing its own network. When the belt-driven starter generator (BISG) is operating as a generator, the converter works in the buck mode, the two batteries are charged and the power is supplied to the loads. When the BISG operates as a motor, the DC/DC converter operates in the boost mode and the power flows from the 12 V side to the 48 V side [5]. Some manufacturers include a starter on the low voltage side in order to start the motor in case of failure of the 48 V network.

3. Modeling of MHEV DC-DC converter

The black-box modeling methodology presented in this paper is based on a LSTM-NN. This is a type of recursive neural network commonly applied in time-series related problems [25]. It uses the actual and predicted values at the input and output terminals of the power converter to predict the future output values. This section presents a general overview of the well-known LSTM-NN architecture, next a specific LSTM-NN topology for the analyzed problem is presented, and finally, a strategy to tune the hyper-parameters is developed.

3.1. Long short term memory neural networks (LSTM-NN)

Traditional recurrent neural networks (RNN) are applied for

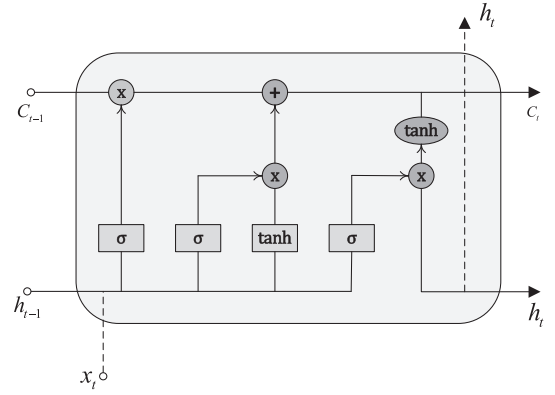


Fig. 3. LSTM cell unit.

modeling time-series problems with good accuracy. However, their main problem is the difficulty of learning long term dependencies of the data because of the gradient decay. LSTM-NNs overcome this problem [26]. This type of network is able of discarding unrelated information during the training process, leading to the preservation of the back-propagation error over time. The weights of the neural network are updated based on the memory units of the LSTM cells, which are not affected by external perturbations [27]. A great advantage of LSTM-NNs, compared to traditional RNNs, is that they keep learning at every time step, even in case of numerous time steps.

The fundamental unit of the LSTM-NN is the cell, which includes memory blocks that store, write, read and erase information during the training process. This is accomplished thanks to the presence of gates, that allow the network to update the weights of the neurons at each iteration [28].

Fig. 3 shows the architecture of LSTM cells, where c refers to the cell memory state, h is the hidden state, x is the input value and σ represents the sigmoid function. Subscript t refers to the time step of the cell, since there are as many cells as time steps in the time series. The LSTM is a block where different cells are connected sequentially, so that each cell generates a future output depending on the past values. Fig. 3 shows the input, forget, output and input modulation gates of a LSTM cell. The first three are controlled by the sigmoid function σ , and they generate an output between 0 and 1 based on the significance of the h_{t-1} input value [28], whereas the input modulation gate uses the 'tanh' activation function. Pointwise operations are applied inside the cell in order to connect the four gates and to obtain the output values. The interaction between the gates and the capacity of forgetting irrelevant information allows the network learning long range temporal dependencies [9]. It is desirable for the problem studied in this paper, because the outputs of a DC-DC converter depend on the previous states [29].

This architecture allows calculating the output values just by applying mathematical calculations to the input data. Eq. (1) presents the operations that take place at the different gates of the LSTM cell.

$$\begin{aligned} i &= S(W_i x_t + R_i h_{t-1} + b_i) \\ f &= S(W_f x_t + R_f h_{t-1} + b_f) \\ o &= S(W_o x_t + R_o h_{t-1} + b_o) \\ g &= \tanh(W_g x_t + R_g h_{t-1} + b_g) \end{aligned} \quad (1)$$

where W refers to the input weights, R to the recurrent weights, b to the bias, the subscript to the respective gate and the $S()$ function represents the sigmoid function applied to each gate. Then, the cell memory state and the hidden state are calculated as follows,

$$c_t = f \cdot c_{t-1} + i \cdot g \quad (2)$$

$$h_t = o \cdot \tanh(c_t) \quad (3)$$

Finally, by using the updated hidden states, it is possible to calculate the predicted output values as,

$$y_i = W_{fc}h_i + b_{fc} \quad (4)$$

where W_{fc} and b_{fc} are the weights and bias of the fully-connected layer, respectively.

3.2. Training of the LSTM-NN

To define the characteristics of the LSTM-NN and the tuning strategy, it is necessary to establish the inputs and outputs of the black-box model. Considering that the only variables that can be measured in real situations are the inputs and outputs of the DC-DC bidirectional converter, the selected inputs of the black-box are the voltage and current at the 48 V side, while the outputs are the voltage and current at the 12 V side. However, the inputs and outputs might be in the opposite way if the 48 V is defined as the output.

After understanding the nature of MHEVs bidirectional converters, the working principle of the LSTM-NN and defining the inputs and outputs of the system, the next step is to choose the principal characteristics of the LSTM-NN in order to model accurately the behavior of the DC-DC converter. According to the LSTM-NN topologies presented in [30], and given the complexity of the analyzed dataset, just one hidden layer is required to obtain precise results after the training process. When adding extra layers to this type of NN, the training process demands more computational resources.

The process of tuning the hyper-parameters is done by means of a Bayesian optimization algorithm (BOA). It finds an optimal combination of hyper-parameters by training a small number of NNs. The algorithm trains the NN with an initial set of hyper-parameters, and based on the points close to the minimum, the hyper-parameters are changed at each iteration [31]. The objective function is defined as,

$$x^* = \underset{x \in X}{\operatorname{argmax}} f(x) \quad (5)$$

where $f(x)$ is a surrogate function generated by sampled points, x^* refers to the optimal set of hyper-parameters and X to the space of possible values. Given the surrogate function, the idea is to reduce the uncertainty of the model by evaluating different LSTM-NNs with different sets of hyper-parameters. A good balance between exploration and exploitation of the candidate points is needed in the optimization process in order to have a high prediction accuracy with small uncertainty [32].

Since the problem studied in this paper uses a large dataset, it is not efficient to tune several hyper-parameters. Therefore, just four of them were chosen, based on how they affect the training process and model accuracy. These hyper-parameters are the hidden layer neurons, learning rate, gradient decay factor (GDF) and the L_2 regularization value. The first one determines the number of interconnections within the NN and its complexity, the second one refers to the size of the step that the training algorithm takes in each iteration. The last two hyper-parameters are associated to the solver, which in this case is the 'adam'. This optimizer is meant to solve a non-convex optimization problem that updates the weights of the neurons at each iteration in order to find a local minimum point [33]. The problem is unconstrained and the convergence is affected by the selected hyper-parameters. The GDF represents the decay rate and the L_2 regularization parameter is the weight decay of the training process. These variables help the network to avoid overfitting.

4. Experimental setup and validation

The experimental setup pursues replicating the voltage conversion stage that occurs in MHEVs. To this end, the setup shown in Fig. 2 was implemented in the laboratory. It includes three main elements, the 48 V battery, the DC-DC bidirectional power converter and the 12 V battery. The objective is to obtain a rich dataset based on the measurements at the input and output terminals of the converter, allowing an accurate

Table 1
Lm5170-Q1 Converter Specifications.

Parameter	Value
Input voltage range	6 V – 75 V
Typical input voltage	48 V
Output voltage range	3 V – 48 V
Typical output voltage	12 V
Maximum power	720 W
Switching frequency	100 kHz
Peak efficiency	> 97 %

Table 2
Battery Specifications.

Parameter	48 V Battery	12 V Battery
Nominal discharge current	6.28 A	Not specified
Maximum discharge current	50 A	440 A
Nominal charge current	6.28 A	Not specified
Maximum charge current	12 A	Not specified
Rated capacity	31.9 Ah	64 Ah

training of the LSTM-NN.

To generate the aforementioned dataset, it is important to follow the steps of the proposed methodology, as shown in Fig. 1. Therefore, the first step of the experimental stage is to choose an adequate set of batteries and a proper converter. These elements were selected considering the characteristics, ratings and specifications of the elements that can be found in commercial MHEVs [23]. The chosen converter was the LM5170-Q1 48 V -to- 12 V DC-DC buck-boost converter manufactured by Texas Instruments. This converter is suitable for real MHEV applications and it can be controlled externally by means of its current or a voltage source. The evaluation module of the converter was used to perform the experiments, since it allows the user to modify different parameters based on the required measurements. The bidirectional converter has an inner current control loop and an outer voltage control loop. The first one receives two external signals that determine the direction and magnitude of the current flowing through the converter. It uses a Type-II compensator to stabilize the loop for the buck and boost operations. The outer voltage loop is controlled by a digital circuit and regulates the output voltage of the converter. Both loops are interconnected, since the output error signal of the voltage loop is an input signal for the current loop. Table 1 shows the characteristics of the converter.

The MD-BB0 Li-Ion 48 V battery pack from Torrot was used, which is

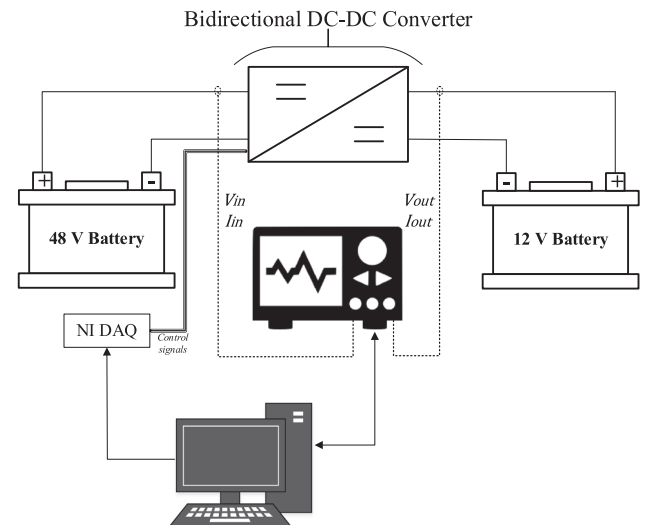


Fig. 4. Experimental setup.

intended for automotive applications. Despite it is not specifically designed for MHEVs, the characteristics shown in Table 2 are quite similar to the 48 V batteries of commercially available MHEVs. On the other hand, the T4 12 V lead acid battery from Augi was used, which is widely applied in vehicles.

Table 2 summarizes the specifications of the batteries dealt with in this work.

After selecting the equipment, the following step is to define an experimental setup based on the type of data needed to train and test the LSTM-NN. As already mentioned, the training stage of the neural network requires a rich dataset that includes different operating points and transient responses of the bidirectional converter. Thus, several conditions must be measured based on the maximum rating and response time of the converter. An automatic acquisition system was implemented in order to collect the training and test datasets in a fast and efficient way.

Fig. 4 shows the schematic of the setup implemented in the laboratory where the oscilloscope and the bidirectional converter are centrally controlled by a computer and a DAQ device. A Python code synchronizes the control signals of the converter and the data acquisition system. This code sends a current command to the control loops of the bidirectional converter. Based on this value, it adjusts the measurement settings of the oscilloscope (trigger, coupling and time scale). The maximum and minimum values of the current command are fixed based on the power rating of the batteries and the converter. The number of iterations is defined by the user and indicates the number of experiments of the dataset. Each experiment contains five signals: time, input voltage, input current, output voltage and output current of the bidirectional converter. It is not necessary to connect loads to any of the batteries, as the converter provides the direction and magnitude of the current.

It is necessary to pre-process the input data in order to filter the noise and to avoid overfitting, thus increasing the training speed of the neural network. The pre-processing consists of an interpolation and the application of a simple moving average filter. First, the interpolation is applied to reduce the number of points of each experiment of the dataset. Next, the moving average filter is applied to eliminate the high-frequency noise from the measurements. The number of points taken in the moving average is chosen in order to keep the main characteristics of the transient response. The last step of this stage is to divide the processed data set into the training, validation and test sets. This partition is done based on the information available on the dataset, because it directly affects the results of the training of the LSTM-NN. The same pre-processing strategy must be applied to these three subsets.

4.1. Validation

The 48 V-to-12 V DC-DC bidirectional converter model is validated by the test data set to predict the output values provided by the LSTM-NN, which are compared against the actual values, and the error between both signals is calculated.

After pre-processing the raw acquired experimental data, the dataset was partitioned in three different subsets as shown in Fig. 1. The third set, i.e., the test dataset, is used to validate the LSTM-NN by testing the obtained model under conditions other than those of training. This is done to verify the robustness of the model and that the trained neural network does not have overfitting issues. The accuracy of the predictions is calculated by means of the root mean squared error and the coefficient of determination, which are two indicators used to evaluate modeling and fitting problems [30]. RMSE values as low as possible and R^2 values close to 1 are desirable.

5. Experimental results

This section presents the modeling results of a real DC-DC buck-boost converter used in MHEVs based on experimental data. The results obtained are compared with three black-box methods found in the

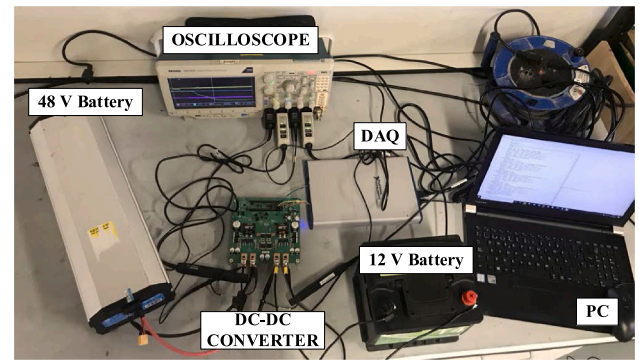


Fig. 5. Experimental setup of the MHEV electrical system.

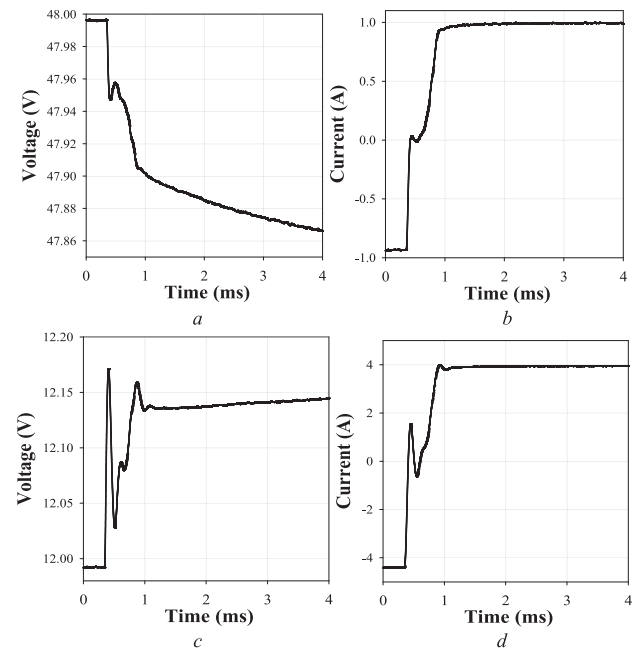


Fig. 6. Converter measurements per experiment. (a) Input voltage, (b) Input current, (c) Output voltage and (d) Output current.

technical literature, that is, NARX neural networks [17], polytopic modeling [34] and the state-space averaging method [19,35]. These are methods that show accurate results in identifying the dynamics of DC-DC converters and are comparable to the model proposed in this paper, since they require offline measurements. It is noted that a basic topology of a buck-boost converter was modeled in case of the state-space average model.

Fig. 5 shows the experimental setup implemented in the laboratory, from which the experimental data was acquired. This setup has the same configuration as that in Fig. 4.

The equipment used to acquire the data consists of a 4 channel oscilloscope (Tektronix MDO3024 200 MHz 2.5 GS/s; Tektronix, Beaverton, OR, USA), two high-frequency current probes (Tektronix TCP0030A 0.001–20 A 120 MHz; Tektronix, Beaverton, OR, USA) and two high-frequency voltage probes (Tektronix THDP200; Tektronix, Beaverton, OR, USA). A total of 1000 different experiments were performed by controlling the current flowing through the power converter. The sampling frequency of the oscilloscope was set to 2.5 MHz with a length of 10^5 points, resulting in a period of 4 ms. The pre-processing consisted of reducing the number of points of every experiment to 5000 and then applying a moving average filter comprising 10 points.

Fig. 6 presents an example of the four signals acquired in each

Table 3
Bayesian Optimization Range.

Hyper-parameter	Minimum	Maximum
Neurons	5	150
Learning rate (LR)	0.001	0.1
Gradient decay factor (GDF)	0.8	0.99
L_2 regularization value	1e-10	1e-2

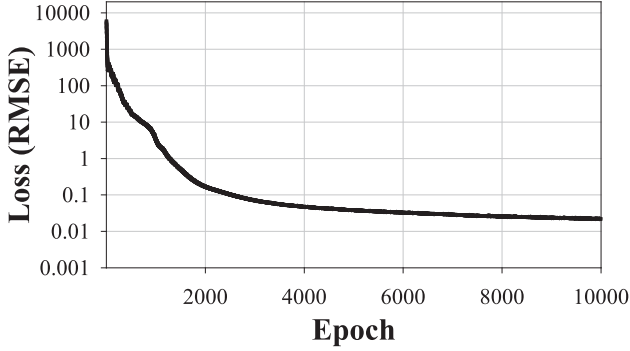


Fig. 7. Training progress.

Table 4
Results.

Method	RMSE		R^2		Time elapsed
	I_{in}	V_{out}	I_{in}	V_{out}	
LSTM-NN	0.1204	0.0023	0.9995	0.9998	274 s
NARX-NN [17]	1.4043	0.0067	0.9761	0.9877	12458 s
Polytopic model [34]	1.3740	0.0126	0.9522	0.9565	26 s
State-space averaging method [19,35]	1.4410	0.0222	0.9465	0.9539	312 s

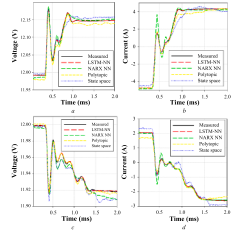


Fig. 8. Transient response measurement and estimation of (a) Output voltage (boost mode to buck mode), (b) Output current (boost mode to buck mode), (c) Output voltage (buck mode to boost mode), (d) Output current (buck mode to boost mode).

experiment. In this case it is well appreciated the transient response of the bidirectional converter.

A total of 750 experiments was assigned to the training dataset, whereas the validation and test datasets contain 125 experiments each. The next step is to tune the hyper-parameters of the NN.

Table 3 presents the minimum and maximum bounds of the variables to be tuned by the Bayesian optimization algorithm. A total number of 30 NNs were trained by the BOA, each training process run for 100 epochs.

The BOA took 12 min to train the 30 neural networks. It was found that the optimal point has the following hyper-parameters: 29 neurons, LR = 0.03426, GDF = 0.974 and L_2 = 0.00169. Subsequently, the LSTM-NN was trained with these hyper-parameters for 1000 epochs using the 'adam' solver.

Fig. 7 shows the evolution of the training progress.

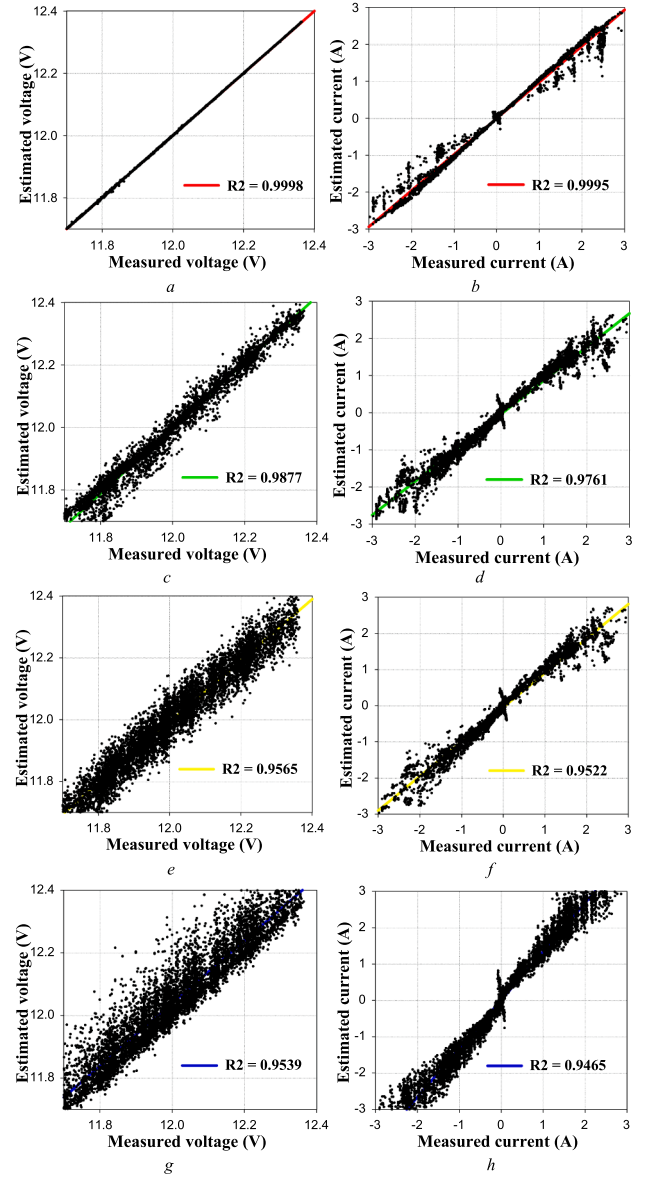


Fig. 9. Scatter plot comparing measured and estimated signals. (a) output voltage (LSTM-NN), (b) output current (LSTM-NN), (c) output voltage (NARX), (d) output current (NARX), (e) output voltage (Polytopic), (f) output current (Polytopic), (g) output voltage (State space), (h) output current (State space).

The following step consists of validating the NN when tested under different conditions than the ones used during the training stage.

Table 4 shows the RMSE and R^2 values obtained after evaluating 125 experiments from the test dataset. It also compares these values to the ones obtained by means of the predictions made using the other three methods. The proposed methodology outperforms the other three techniques because the RMSE is lower and the R^2 coefficient is closer to 1.

To better understand the accuracy of the proposed model, Fig. shows the actual and predicted outputs of two experiments of the test dataset. Fig. a) and b) show an experiment in which the converter changes its operating mode from boost to buck, whereas in Fig. c) and d) the operating mode changes in the opposite way. In both cases, the estimation of the LSTM-NN is close to the measurements. Fig. 8 also show that the accuracy of the transient response is better than those estimated by the other three models. The LSTM-NN model accurately replicates the steady state and transient response values, while the other methods exhibit some discrepancies. It is worthy to note that the estimation of the

current at the 12 V side of the power converter is slightly less accurate than the estimation of the voltage. This might be because the transient response of the output current is wider than the one for the voltage, depending on the experiment.

Fig. 9 plot the estimation done by the four methods against the experimental measurements for different operating. Such results prove the robustness of the proposed model since the slope is very close to unity. The proposed method outperforms the other three approaches, since the dispersion of the points is much smaller and the coefficients of determination are greater.

The results presented in this section proved the effectiveness of the modeling approach proposed in the paper, as well as the accurateness in comparison with other methods found in the technical literature.

6. Conclusion

In this paper, a methodology for black-box modeling of a 48 V –to- 12 V DC-DC bidirectional converter has been presented. The use of a deep learning algorithm enables the model to reproduce the behavior of a nonlinear switching device, such as the bidirectional converter used in MHEVs. Based on the converter characteristics, a hyper-parameter tuning algorithm and a specific topology of the LSTM-NN have been discussed. An adequate experimental setup was needed to generate the datasets to be used in the training process of the LSTM-NN. Both steady state and transient response data of the converter were identified with the same NN. The proposed model has a high accuracy when testing new operating conditions different from those used during the training stage, while estimating the output voltage and current with low RMSE values and high coefficients of determination. The values of these indicators prove the robustness of the proposed model, which is able to replicate the nonlinearities of the bidirectional converter under different operating points. It has also been shown that the trained LSTM-NN outperforms three estimation methods commonly used to model power converters in the technical literature. Finally, the proposed approach can also be applied for an accurate black-box modeling of many other power electronics devices.

CRediT authorship contribution statement

Gabriel Rojas-Dueñas: Conceptualization, Software, Writing – original draft, Validation. **Jordi Roger Riba:** Conceptualization, Resources, Writing – review & editing, Supervision. **Manuel Moreno-Eguilaz:** Conceptualization, Supervision, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Acknowledgement

This work has been partially supported by the Generalitat de Catalunya under the project 2017SGR0967, and the European Commission through the Clean Sky program, under the project 755332-AEMSIdFit.

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